

Experimental Setup

Can all physical wave and memory sectors share a stable geometry in the declared reciprocal routing circuit and exact backgrounds?

Experiment: gross.full-spectrum.v1 Model: signal-space.ss-ops-1.six-gate.v1
Runtime configuration: fixtures/research/gross-test-04.json
Plan lock: 8ae4badcb81f956f753711092bdbaf0ee38e330365edb93357eb78e13f797f0a

MODEL AND NUMERICAL SETUP

Action: $SS\ OPS\ 1\ H=(\pi/2)|w\ dagger\ (a-b)|^2$, $\lambda=0$. Three onsite/bond pairs per cycle, two wave spinors and six independent memories per cell. Explicit new incidence; not the Test 3 reduced stencil.
Units: $\ell=\tau_c=1$; $q=k\ell$; $\omega\tau_c$; growth log modulus per cycle; event parameter is not proper time.

Domain: Vacuum orthogonal memories, equal nonzero orthogonal memories, counterport collinear memories. Twenty physical real dimensions. 26 signed directions, four radii .02,.04,.08,.16; three 65-point full-zone paths.

Boundaries: Homogeneous infinite-lattice Bloch matrices; local impulse for three cycles on $9^3/13^3$ periodic boxes before wraparound. No continuum PDE discretization.

Initial Data: Exact analytical physical period-one backgrounds $a=b=0$; $a=b=(.5,0)$; $a=-b=(.5,0)$. Counter memory representatives return in four cycles but projectors in one. PCG64 off-background gate controls.

Solver: Exact reciprocal event solution; full physical Jacobians via centered $h=2e-5, 1e-5$; independent DOP853 with $rtol2e-12\ atol2e-14\ maxstep.05$. Compose twenty-dimensional maps with bond Fourier phase. Save all eigenmodes in analysis.

Tolerances: Conservation and physical recurrence $<1e-10$; differential/reference $<2e-6$; exponential growth fail above $\max(1e-5, 5\text{ matched eigenvalue refinement})$. Unit moduli leave stability unresolved.

COMPETING PREDICTIONS

common-geometry: Every physical sector shares a stable nondegenerate cone, possibly with a common drift and anisotropy; stationary physical memories plus propagating waves contradict this at the vacuum.

sector-and-material: A signal sector can be geometric while memories have additional response. A successful explanation needs independent constitutive and clock predictions; spectra alone cannot establish it.

background-dependence: Equal nonzero waves still decouple memories; counterport waves activate memory response and may introduce growth or mode dependence.

CONTROLS

reference: Verify complete differential map and conserved quantities; method: Independent DOP853 local Jacobians, exact frozen vacuum Fourier reference, off-background local conservation.

physical-memory: Distinguish gauge redundancy and real backreaction; method: Memory phase null, physical overlap change, full/frozen maps on three exact backgrounds.

resolution: Bound numerical and causal errors; method: Two differential steps, four paired radii, two finite boxes, three-cycle local impulse and Fourier comparison.

PREREGISTERED DECISIONS

coverage (Three backgrounds, 20 physical real dimensions, six local gates, both differential steps, all 104 signed low-q points, three full-zone paths and two impulse boxes are finite and complete.)

Pass: Three backgrounds, 20 physical real dimensions, six local gates, both differential steps, all 104 signed low-q points, three full-zone paths and two impulse boxes are finite and complete.

Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.

Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.

Experimental Setup

Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

periodicity (Wave and projector period-one residual below $1e-10$; raw memory spinor phases need not recur.)

Pass: Wave and projector period-one residual below $1e-10$; raw memory spinor phases need not recur.

Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.

Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.

Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

conservation (Local total wave norm, memory norm, Hamiltonian and matrix charge residuals below $1e-10$ on 24 off-background gates.)

Pass: Local total wave norm, memory norm, Hamiltonian and matrix charge residuals below $1e-10$ on 24 off-background gates.

Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.

Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.

Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

linearization (Fine/coarse full map and independent DOP853 local Jacobian differ by less than $2e-6$.)

Pass: Fine/coarse full map and independent DOP853 local Jacobian differ by less than $2e-6$.

Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.

Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.

Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

vacuum-reference (Vacuum map agrees with independent frozen two-port Fourier construction and memory identity to $2e-6$.)

Pass: Vacuum map agrees with independent frozen two-port Fourier construction and memory identity to $2e-6$.

Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.

Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.

Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

phase-quotient (Memory spinor phase leaves wave outputs and projectors invariant below $1e-10$; normalized orientation change modifies its overlap by more than $1e-6$.)

Pass: Memory spinor phase leaves wave outputs and projectors invariant below $1e-10$; normalized orientation change modifies its overlap by more than $1e-6$.

Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.

Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.

Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

Experimental Setup

causal-domain (Localized tangent impulse stays inside Chebyshev radius t after t cycles (outside amplitude $<1e-10$); common centered region agrees between boxes to $1e-10$; one-cycle FFT agrees to $2e-6$.)
Pass: Localized tangent impulse stays inside Chebyshev radius t after t cycles (outside amplitude $<1e-10$); common centered region agrees between boxes to $1e-10$; one-cycle FFT agrees to $2e-6$.
Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.
Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.
Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

backreaction (Counterport full/frozen map difference exceeds $1e-3$; vacuum and equal-port differences below $2e-6$.)
Pass: Counterport full/frozen map difference exceeds $1e-3$; vacuum and equal-port differences below $2e-6$.
Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.
Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.
Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

vacuum-common-cone (Strong all-sector common nondegenerate cone: fail if physical memory identity persists across every nonzero q with residual $<2e-6$, while wave map differs from identity by $>1e-3$.)
Pass: Strong all-sector common nondegenerate cone: fail if physical memory identity persists across every nonzero q with residual $<2e-6$, while wave map differs from identity by $>1e-3$.
Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.
Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.
Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

counter-stability (Fail for a sampled physical multiplier with log modulus $>\max(1e-5, 5 \text{ times matched step-refinement eigenvalue error})$; otherwise unresolved because finite paths and unit moduli do not exclude Jordan growth or unsampled instability.)
Pass: Fail for a sampled physical multiplier with log modulus $>\max(1e-5, 5 \text{ times matched step-refinement eigenvalue error})$; otherwise unresolved because finite paths and unit moduli do not exclude Jordan growth or unsampled instability.
Fail: Complete evidence contradicts the named criterion; scientific fail is not technical failure.
Unresolved: Missing or nonfinite evidence, failed numerical prerequisite, or stability not settled by finite samples.
Uncertainty: Differential refinement, independent ODE and exact Fourier reference; finite direction coverage and degeneracy retained.

VISUAL QUESTIONS

complete-spectrum [evidentiary]: Do all physical wave and memory variations propagate on one cone?
View: complete-spectrum; observables: Floquet phase, memory participation, growth, local impulse, numerical residual

common-geometry: One healthy nondegenerate shared cone across physical sectors.

sector-and-material: Stationary or differently propagating physical memories, or backreaction-dependent response; clock/material explanation still required.

Controls: reference, physical-memory, resolution; uncertainty: All physical modes retained; degenerate eigenvectors are basis-dependent. Finite sampling is not a stability theorem.

Experimental Setup

Data: reports/<report-id>/plot-data/spectrum.csv; transforms: See saved specifications; sorted phase scatter, prescribed sectors and impulse projection only. No fitted geometry or smoothing.

growth-backreaction [evidentiary]: Does the nonzero background activate memory response, and is exponential growth detected?

View: growth-backreaction; observables: Floquet phase, memory participation, growth, local impulse, numerical residual

common-geometry: One healthy nondegenerate shared cone across physical sectors.

sector-and-material: Stationary or differently propagating physical memories, or backreaction-dependent response; clock/material explanation still required.

Controls: reference, physical-memory, resolution; uncertainty: All physical modes retained; degenerate eigenvectors are basis-dependent. Finite sampling is not a stability theorem.

Data: reports/<report-id>/plot-data/spectrum.csv, reports/<report-id>/plot-data/summary.json; transforms: See saved specifications; sorted phase scatter, prescribed sectors and impulse projection only. No fitted geometry or smoothing.

causal-memory-response [evidentiary]: Where can a local physical memory perturbation influence the circuit?

View: causal-memory-response; observables: Floquet phase, memory participation, growth, local impulse, numerical residual

common-geometry: One healthy nondegenerate shared cone across physical sectors.

sector-and-material: Stationary or differently propagating physical memories, or backreaction-dependent response; clock/material explanation still required.

Controls: reference, physical-memory, resolution; uncertainty: All physical modes retained; degenerate eigenvectors are basis-dependent. Finite sampling is not a stability theorem.

Data: reports/<report-id>/plot-data/impulses.csv; transforms: See saved specifications; sorted phase scatter, prescribed sectors and impulse projection only. No fitted geometry or smoothing.

directional-sectors [evidentiary]: Does this specified routing generate leading drift or only finite-band asymmetry?

View: directional-sectors; observables: Floquet phase, memory participation, growth, local impulse, numerical residual

common-geometry: One healthy nondegenerate shared cone across physical sectors.

sector-and-material: Stationary or differently propagating physical memories, or backreaction-dependent response; clock/material explanation still required.

Controls: reference, physical-memory, resolution; uncertainty: All physical modes retained; degenerate eigenvectors are basis-dependent. Finite sampling is not a stability theorem.

Data: reports/<report-id>/plot-data/directional.csv; transforms: See saved specifications; sorted phase scatter, prescribed sectors and impulse projection only. No fitted geometry or smoothing.

accuracy-audit [evidentiary]: Are recurrence, differential accuracy, phase redundancy and local conservation controlled?

View: accuracy-audit; observables: Floquet phase, memory participation, growth, local impulse, numerical residual

common-geometry: One healthy nondegenerate shared cone across physical sectors.

sector-and-material: Stationary or differently propagating physical memories, or backreaction-dependent response; clock/material explanation still required.

Controls: reference, physical-memory, resolution; uncertainty: All physical modes retained; degenerate eigenvectors are basis-dependent. Finite sampling is not a stability theorem.

Data: reports/<report-id>/plot-data/summary.json; transforms: See saved specifications; sorted phase

Experimental Setup

scatter, prescribed sectors and impulse projection only. No fitted geometry or smoothing.

RESOURCE LIMITS

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{"max_memory_mb": 512, "max_output_mb": 48, "max_wall_seconds": 180}
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PROHIBITED INFERENCES

No emergent spacetime, autonomous clock, physical detector record, particle identity or protected topology.

No generic exclusion of nonzero backgrounds, drift, anisotropy or geometry with derived material response.

No stability proof from unit Floquet moduli; no energy meaning assigned to tangent norm or eigenvector participation.

No equivalence asserted between the six-gate routing ansatz and Test 3 reduced stencil.